

$$\text{I} \\ \text{a)} \quad \lim_{x \rightarrow 0} \frac{\ln(1+x)}{x}$$

$$\text{b)} \quad \text{natura seriei} \quad \sum_{n \geq 1} n \ln \left(1 + \frac{1}{\sqrt{n^2 + 3n}} \right)$$

II punctele de extrem local

$$f: \mathbb{R}^2 \rightarrow \mathbb{R}, \quad f(x, y) = xy^2 + x^2y - x - y - 10$$

III raza de convergență, mulțimea de convergență,
suma seriei de puteri $\sum_{n \geq 1} \frac{x^n}{n \cdot 3^n}$

$$\text{IV} \quad \int_0^1 \int \frac{x}{y^2} dx dy$$

$$D = \left\{ (x, y) \in \mathbb{R}^2 \mid y \geq -x, y \leq \frac{x+12}{2}, x \leq -1 \right\}$$